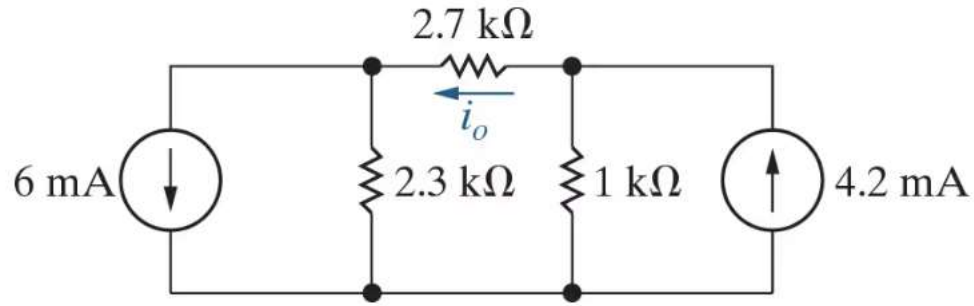
Problems - Week 7

- Source Transformations
- Thevenin and Norton Equivalents

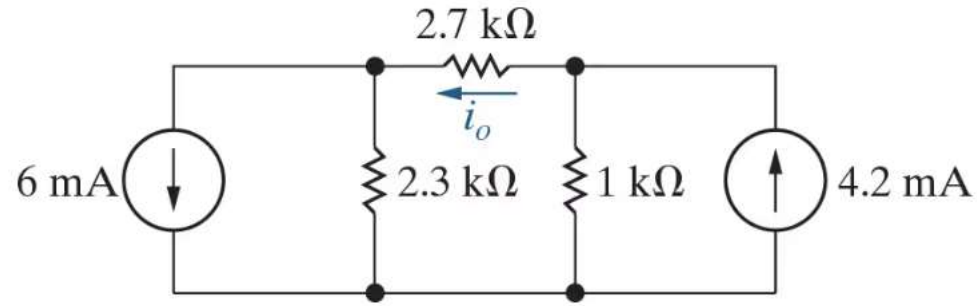
4.59 **a.** Use source transformations to find the current i_o in the circuit in Fig. P4.59.

b. Verify your solution by using the node-voltage method to find i_o.

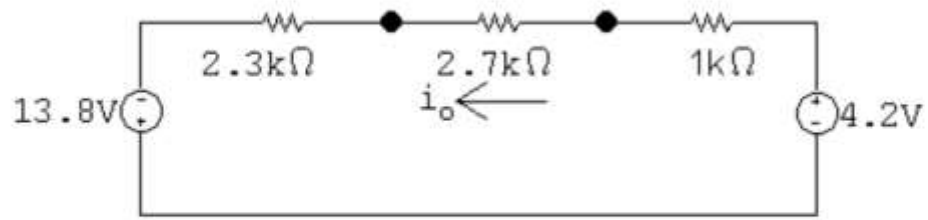


4.59 **a.** Use source transformations to find the current i_o in the circuit in Fig. P4.59.

b. Verify your solution by using the node-voltage method to find i_o.



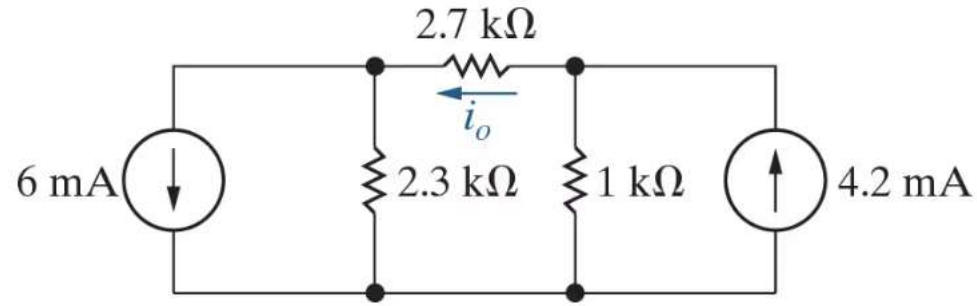
[a] Apply source transformations to both current sources to get



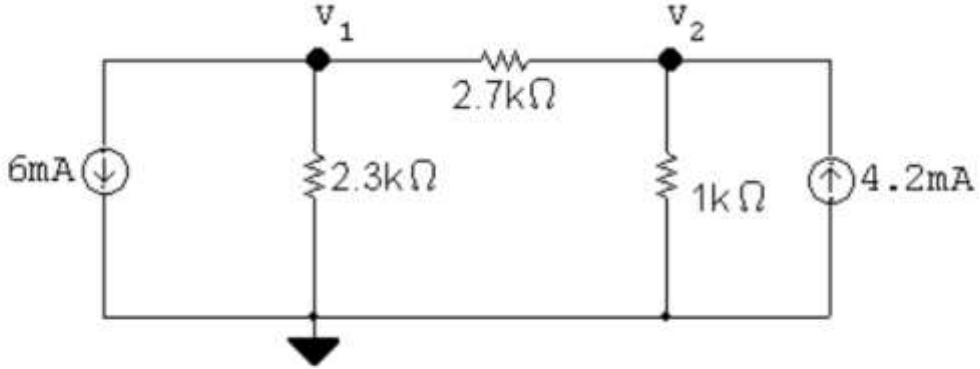
$$i_o = \frac{13.8 + 4.2}{2700 + 2300 + 1000} = 3\,\text{mA}.$$

4.59 **a.** Use source transformations to find the current i_o in the circuit in Fig. P4.59.

b. Verify your solution by using the node-voltage method to find i_o.



[b]



The node voltage equations:

$$6 \times 10^{-3} + \frac{v_1}{2300} + \frac{v_1 - v_2}{2700} = 0;$$

$$\frac{v_2}{1000} + \frac{v_2 - v_1}{2700} - 4.2 \times 10^{-3} = 0.$$

Place these equations in standard form:

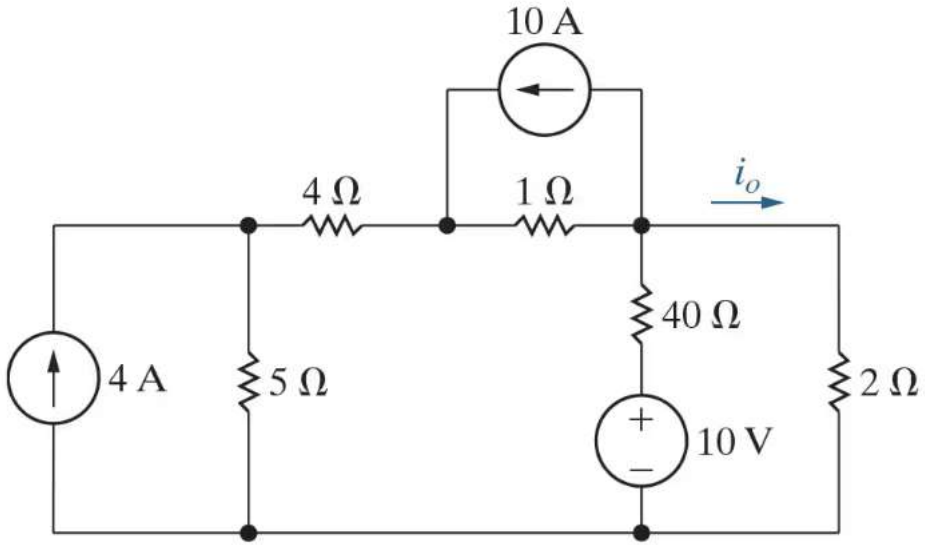
$$v_1\left(\frac{1}{2700} + \frac{1}{2300}\right) + v_2\left(-\frac{1}{2700}\right) = -6 \times 10^{-3};$$

$$v_1\left(-\frac{1}{2700}\right) + v_2\left(\frac{1}{1000} + \frac{1}{2700}\right) = 4.2 \times 10^{-3}.$$

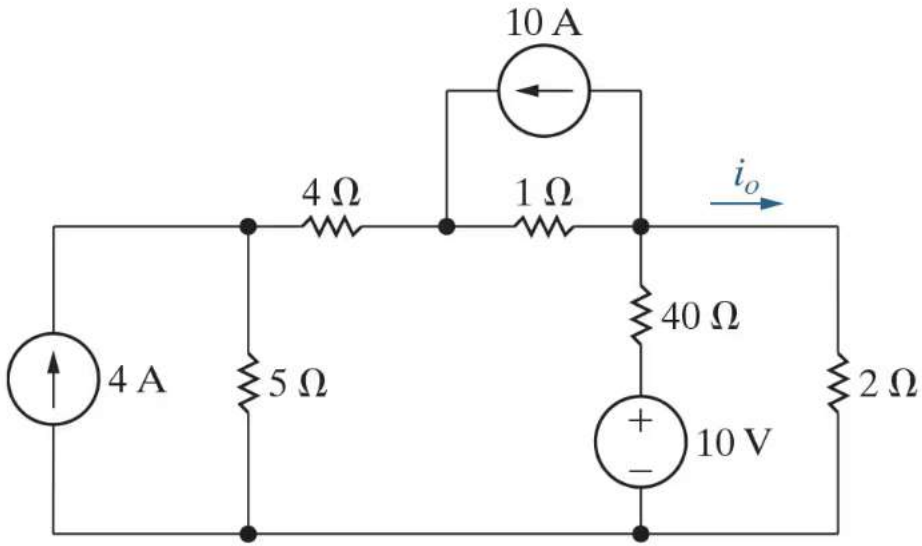
Solving, $v_1 = -6.9$ V; $v_2 = 1.2$ V.

$$\therefore \quad i_o = \frac{v_2 - v_1}{2700} = 3\,\text{mA}.$$

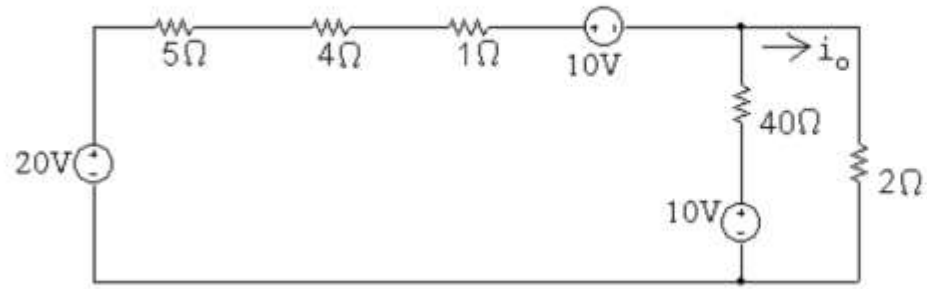
4.60 **a.** Use a series of source transformations to find i_o in the circuit in Fig. P4.60.



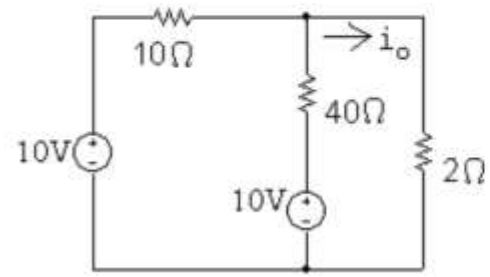
4.60 a. Use a series of source transformations to find i_o in the circuit in Fig. P4.60.



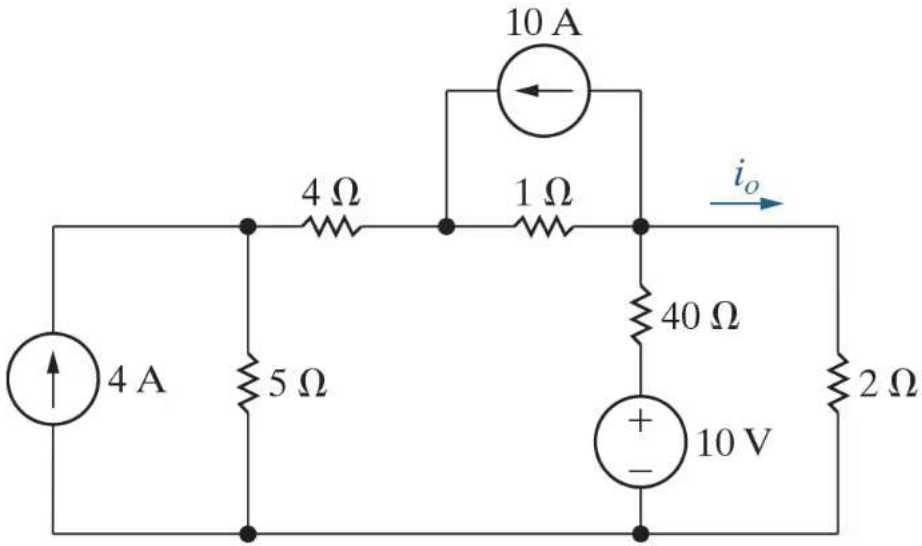
[a] Applying a source transformation to each current source yields

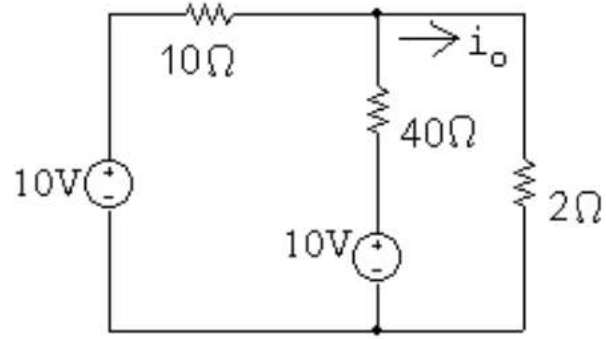


Now combine the 20 V and 10 V sources into a single voltage source and the 5 Ω, 4 Ω and 1 Ω resistors into a single resistor to get

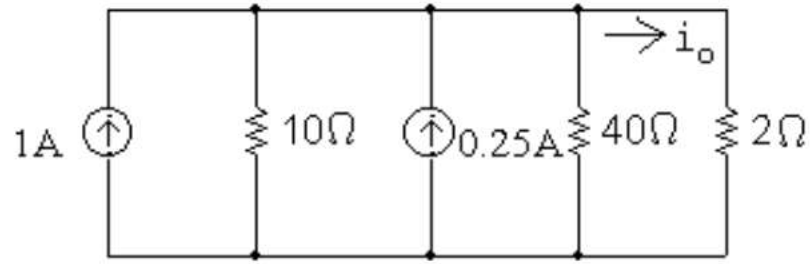


4.60 a. Use a series of source transformations to find i_o in the circuit in Fig. P4.60.

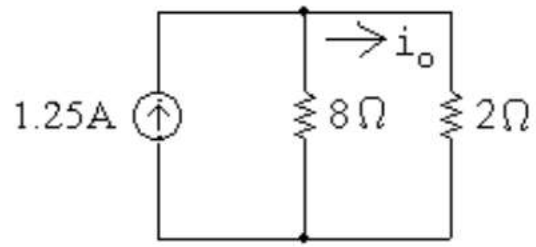




Now use a source transformation on each voltage source, thus

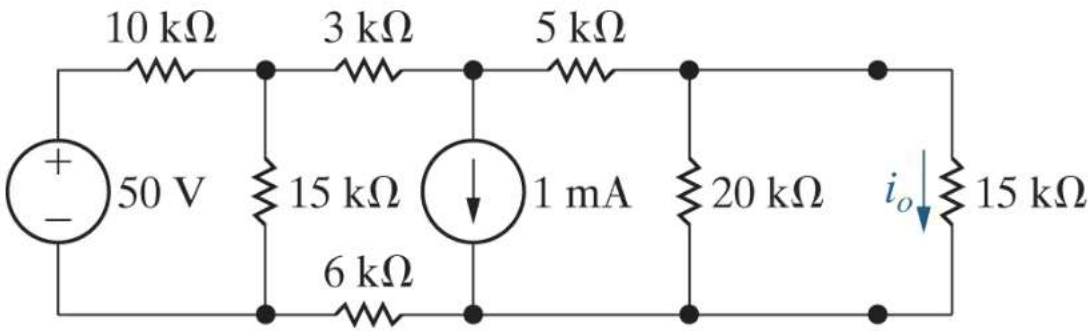


which can be reduced to

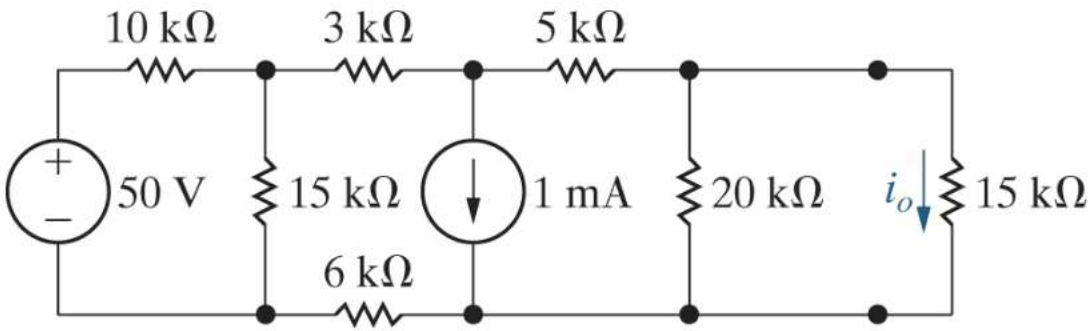


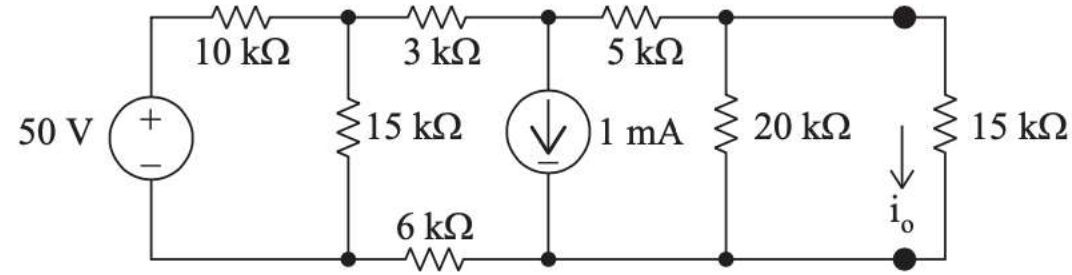
$$\therefore \quad i_o = \frac{2\|8}{2}(1.25) = 1\,\text{A}.$$

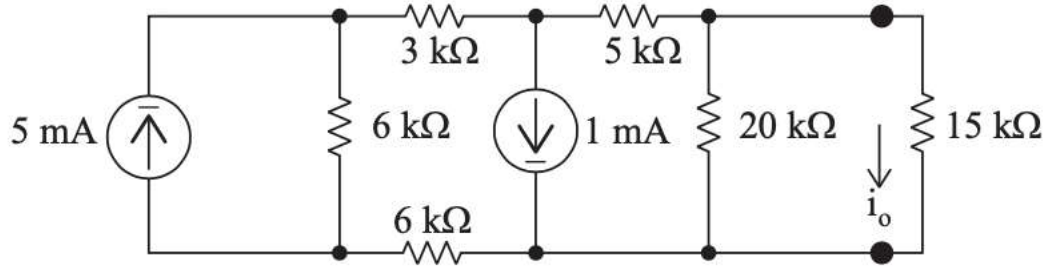
a. Find the current i_o in the circuit in Fig. P4.61 by making a succession of appropriate source transformations.

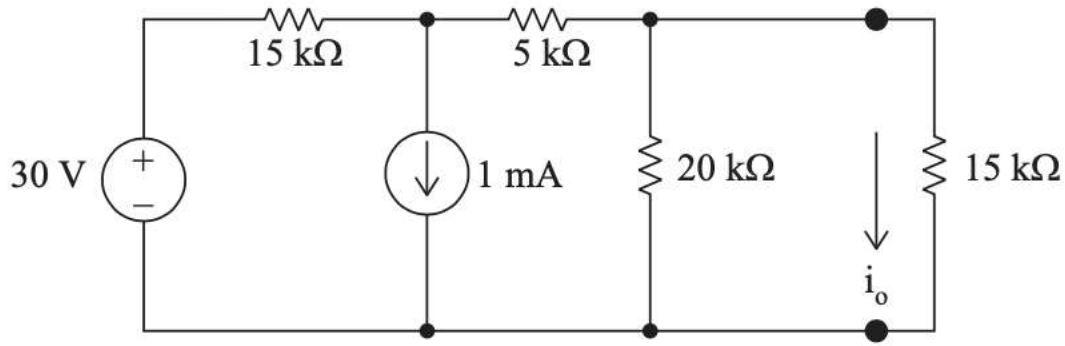


a. Find the current i_o in the circuit in Fig. P4.61 by making a succession of appropriate source transformations.

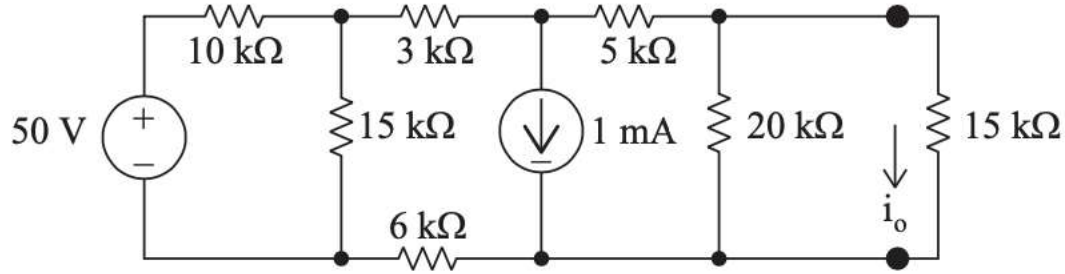


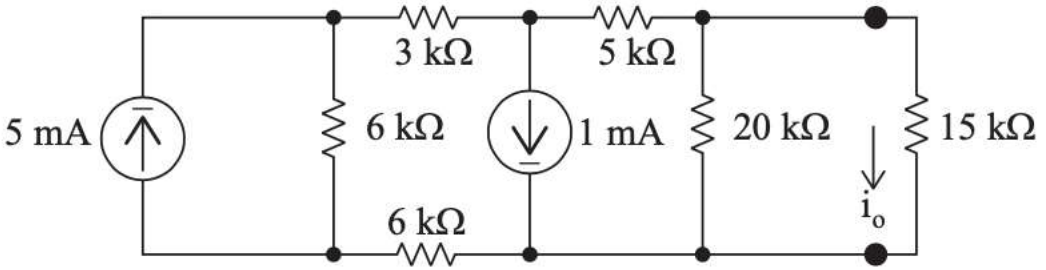


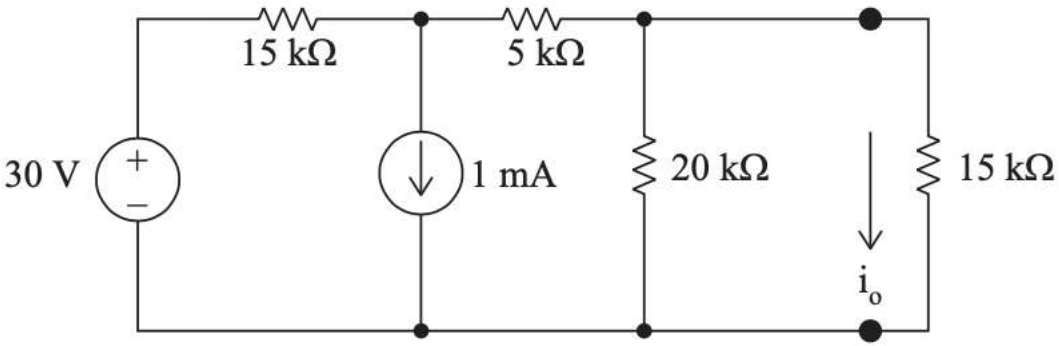


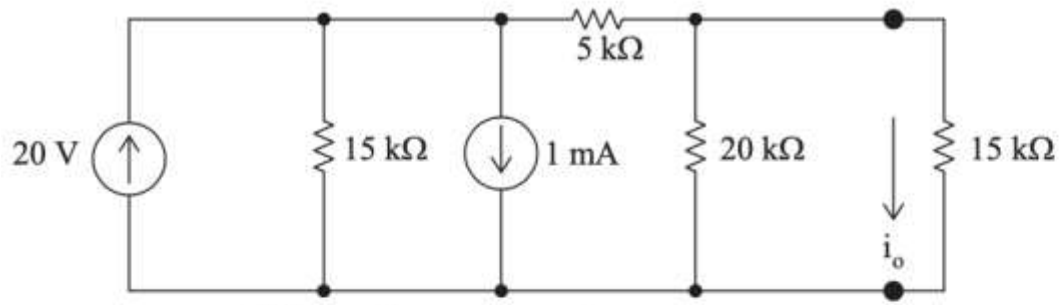


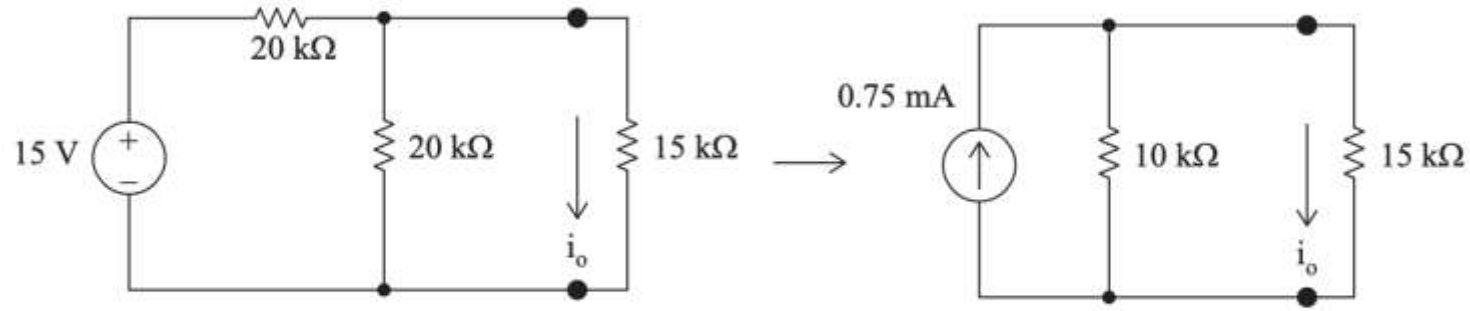
a. Find the current i_o in the circuit in Fig. P4.61 by making a succession of appropriate source transformations.

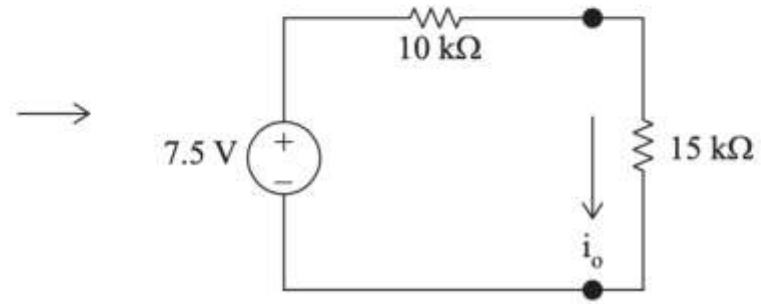




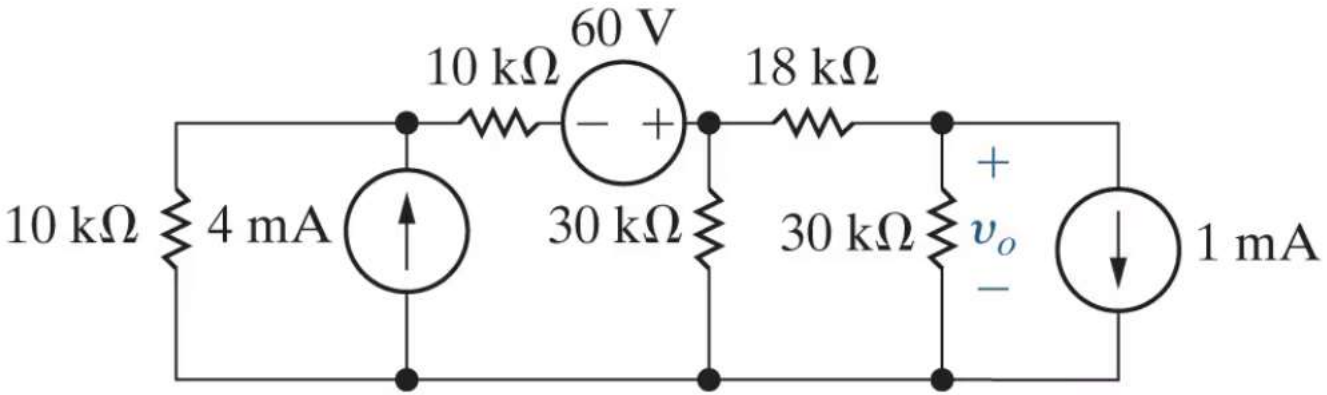




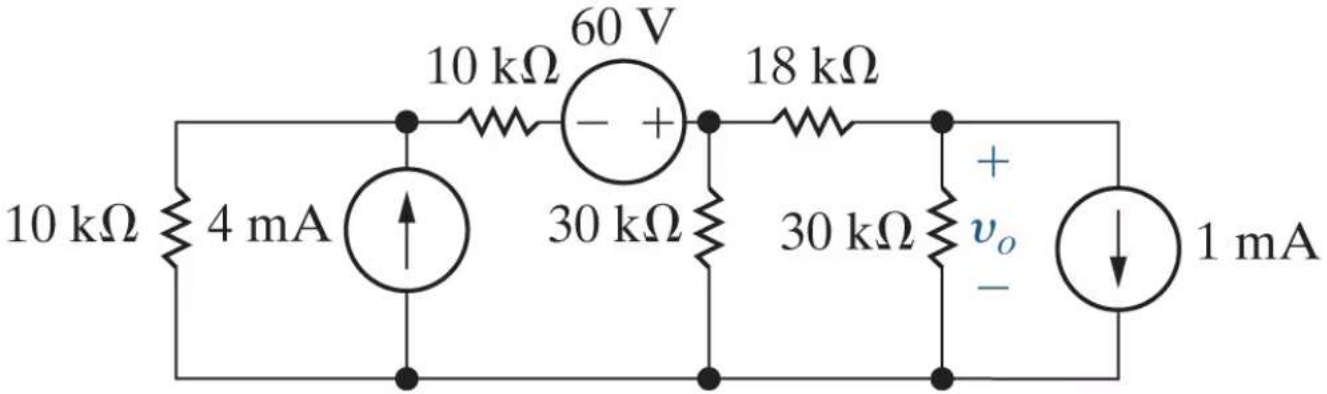


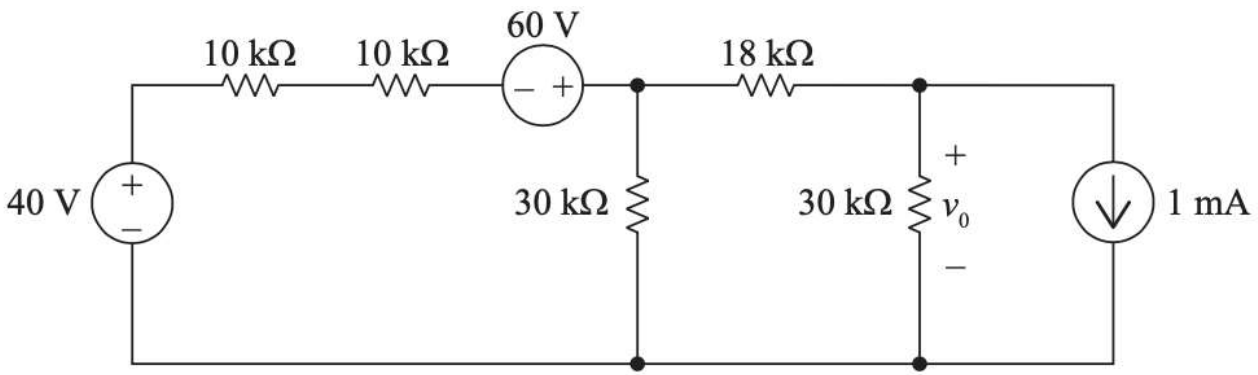


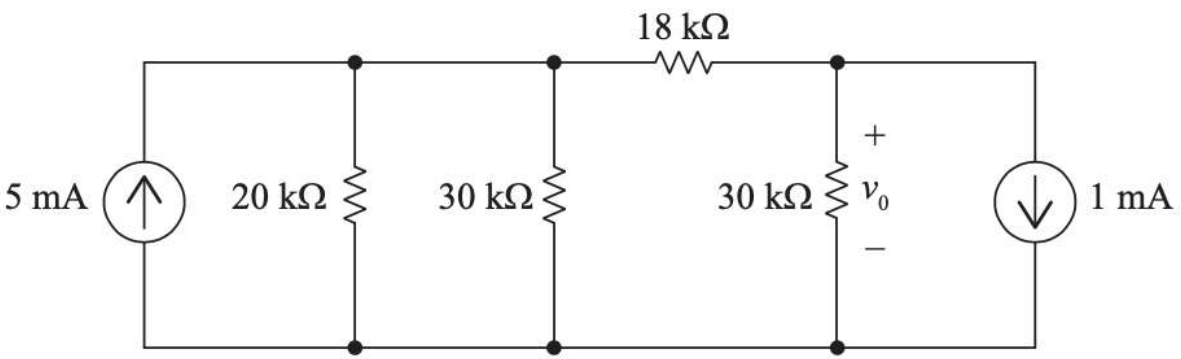
Make a series of source transformations to find the voltage v_o in the circuit in Fig.

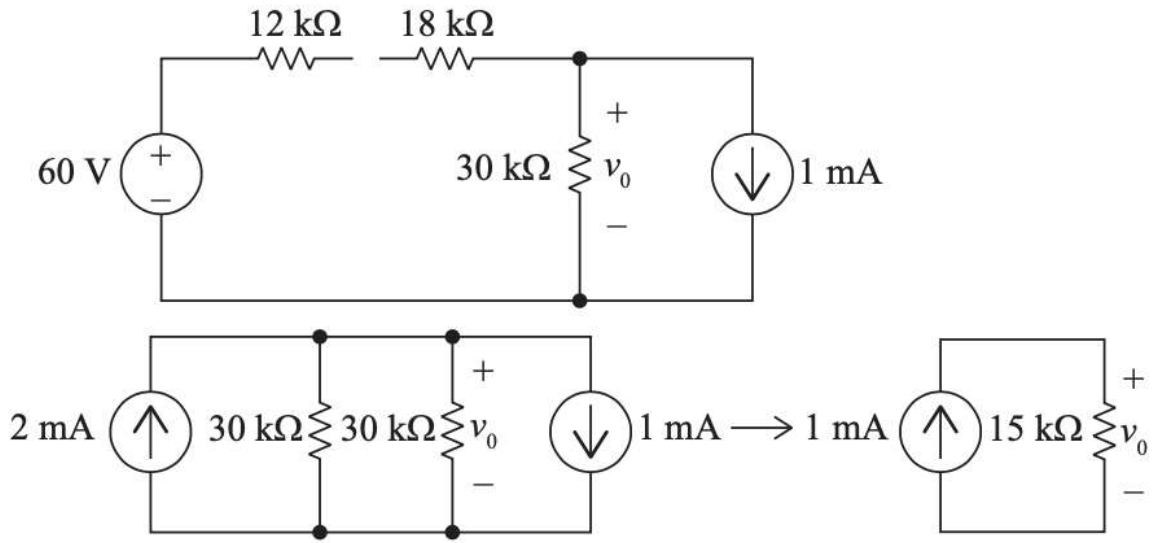


Make a series of source transformations to find the voltage v_o in the circuit in Fig.



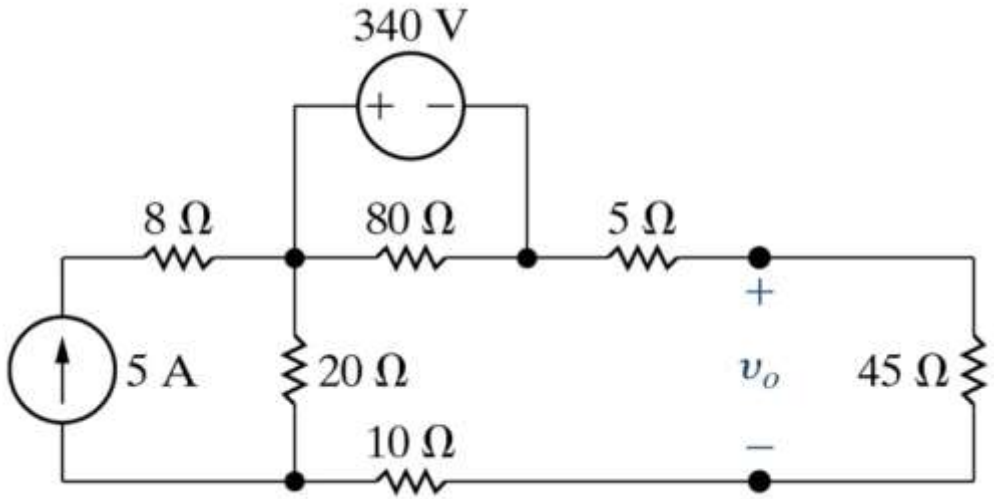




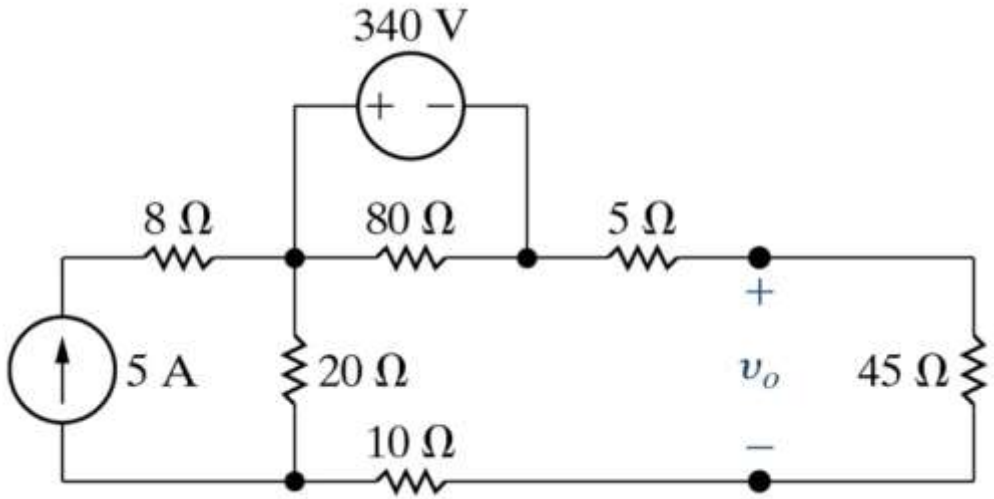


$$v_o = (15{,}000)\,(0.001) = 15 \text{ V}.$$

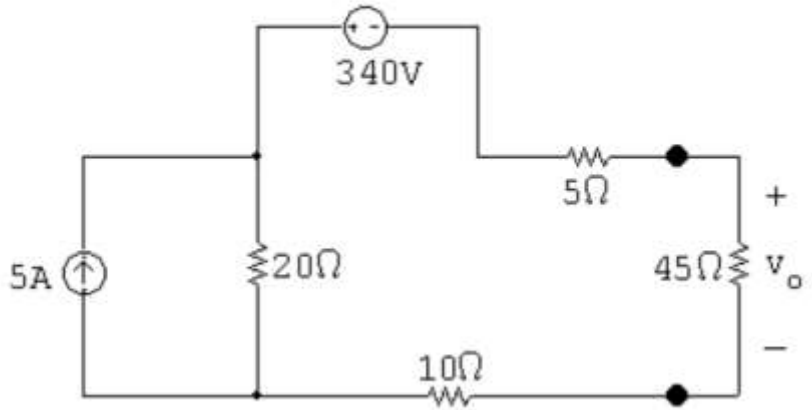
Use source transformations to find v_o in the circuit in Fig. P4.63.



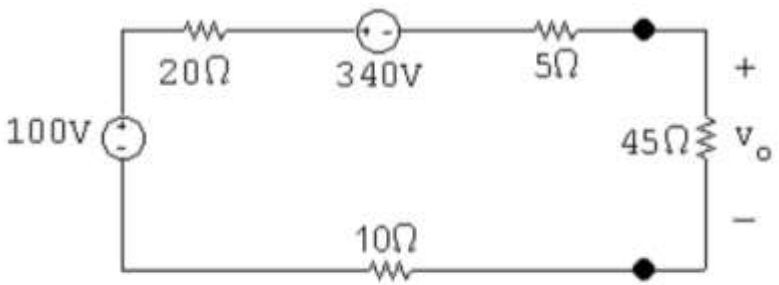
Use source transformations to find v_o in the circuit in Fig. P4.63.



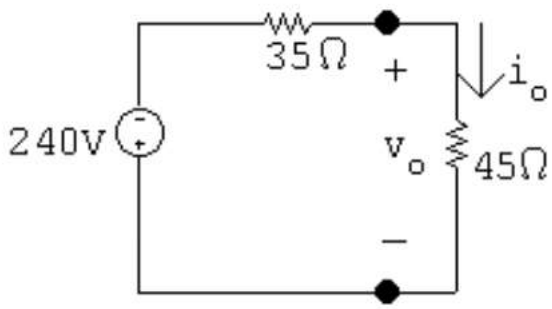
First remove the 8 Ω and 80 Ω resistors:



Next use a source transformation to convert the 5 A current source and 20 Ω resistor:

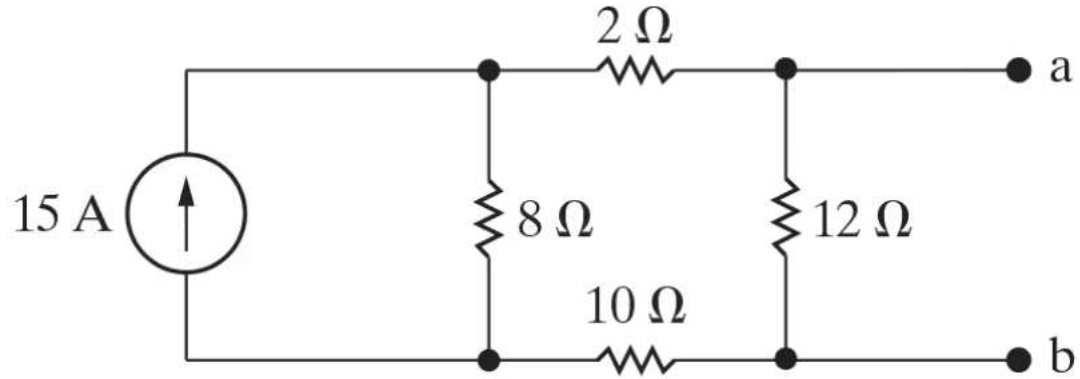


which simplifies to

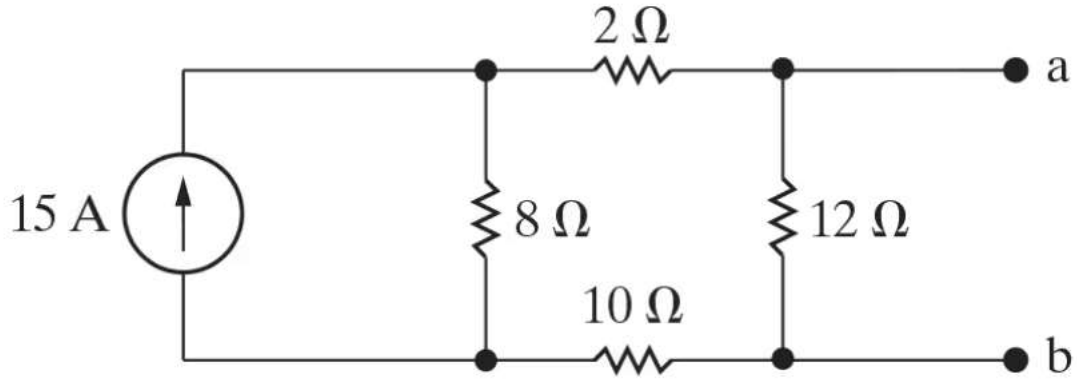


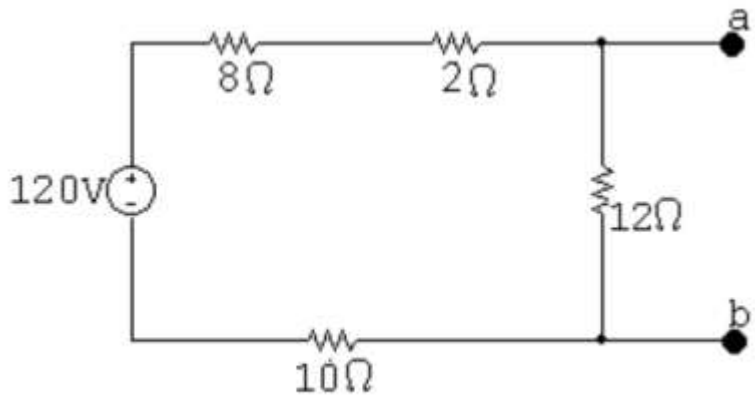
$$\therefore \quad v_o = \frac{45}{80}(-240) = -135\text{V}.$$

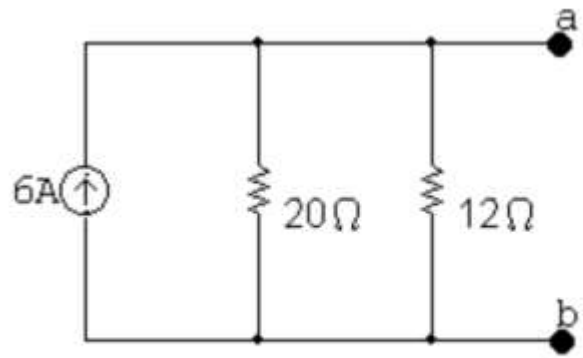
Find the Norton equivalent with respect to the terminals a, b for the circuit in Fig.



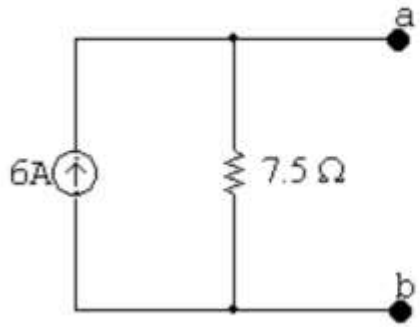
Find the Norton equivalent with respect to the terminals a, b for the circuit in Fig.



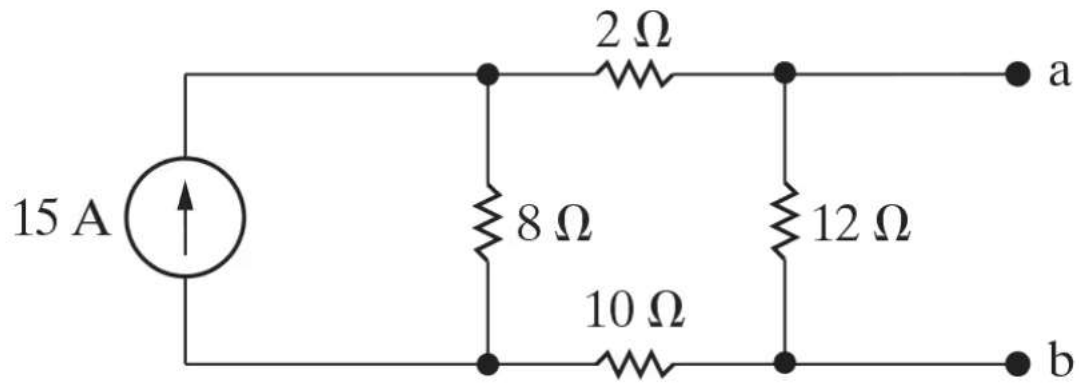




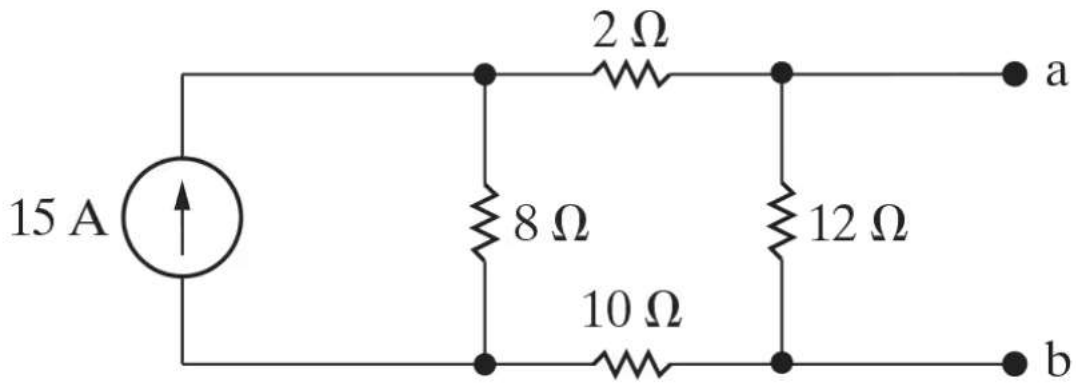
Finally, combine the 20 Ω and 12 Ω parallel resistors to give $R_N = 20 \| 12 = 7.5\,\Omega$ and the Norton equivalent circuit:



a. Find the Thévenin equivalent resistance with respect to the terminals a, b in the circuit in Fig. P4.64 without finding either the open-circuit voltage or the short-circuit current.



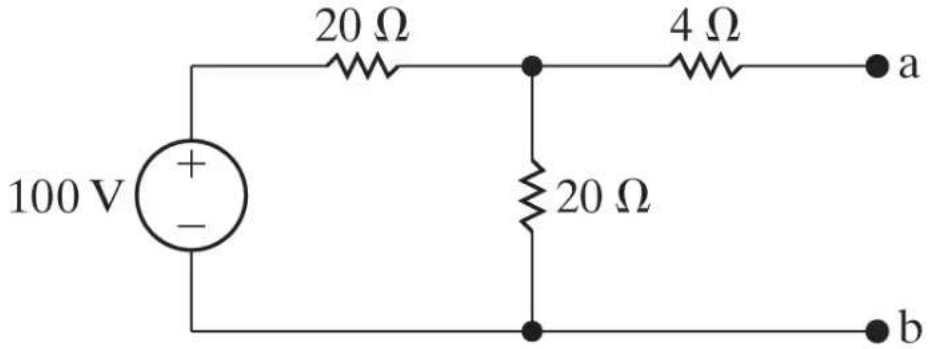
a. Find the Thévenin equivalent resistance with respect to the terminals a, b in the circuit in Fig. P4.64 without finding either the open-circuit voltage or the short-circuit current.



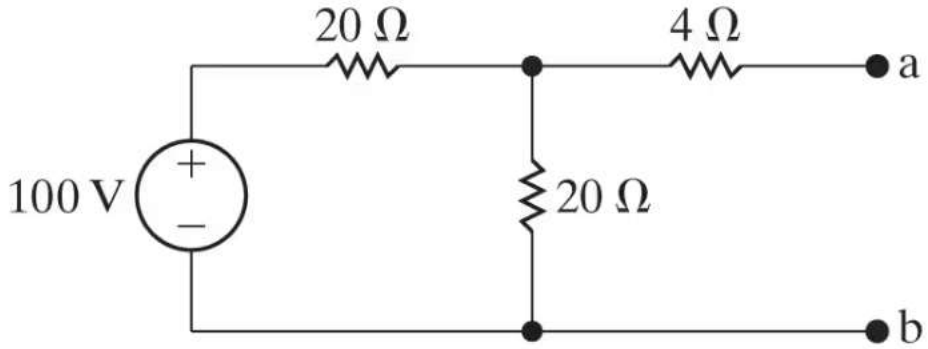
[a] Replace the current source with an open circuit and find the equivalent resistance looking into the terminals a,b:

$$R_{\text{Th}} = 12 \| (2 + 8 + 10) = 12 \| 20 = 7.5\,\Omega.$$

4.65 Find the Norton equivalent with respect to the terminals a and b for the circuit in Fig.



4.65 Find the Norton equivalent with respect to the terminals a and b for the circuit in Fig.

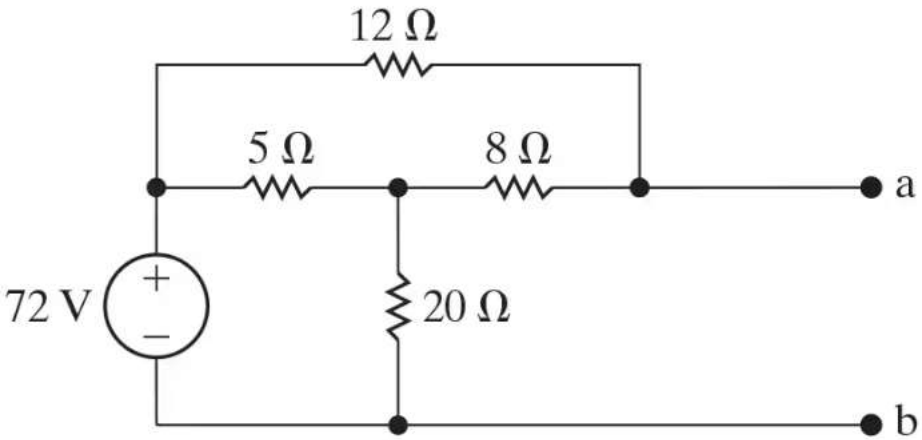


$$R_N = 4 + \frac{20 \times 20}{40} = 14\Omega$$

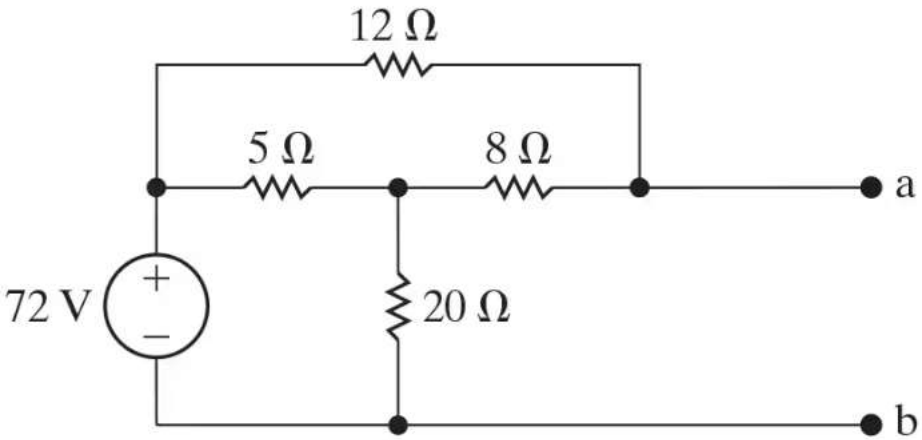
$$V_{\text{Th}} = \frac{100}{40} \times 20 = 50\text{V}$$

$$I_N = \frac{V_{\text{Th}}}{R_N} = 3.57 \text{ A}$$

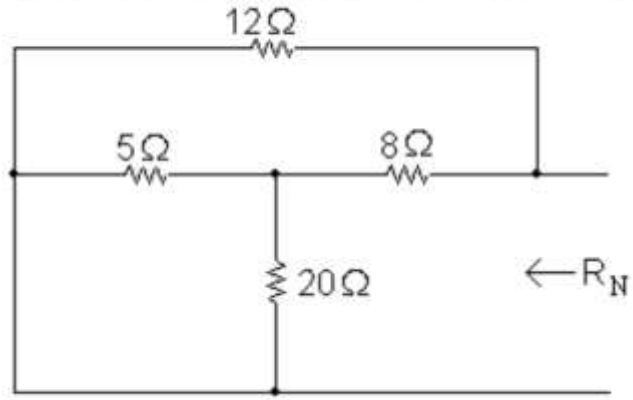
4.66 Find the Norton equivalent with respect to the terminals a, b for the circuit in Fig. P4.66.



4.66 Find the Norton equivalent with respect to the terminals a, b for the circuit in Fig. P4.66.

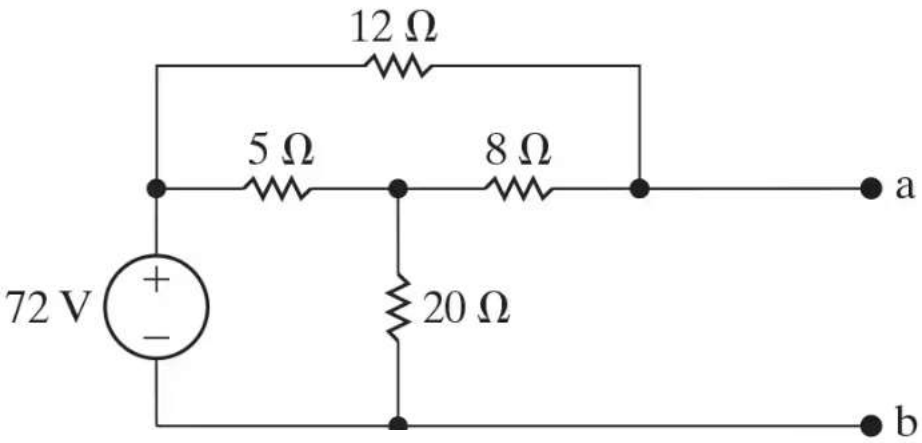


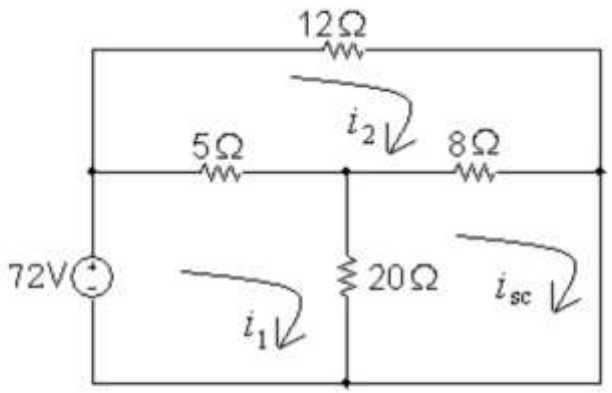
To find R_N, replace the 72 V source with a short circuit:



Note that the 5 Ω and 20 Ω resistors are in parallel, with an equivalent resistance of $5\|20 = 4\,\Omega$. The equivalent 4 Ω resistance is in series with the 8 Ω resistor for an equivalent resistance of $4 + 8 = 12\,\Omega$. Finally, the 12 Ω equivalent resistance is in parallel with the 12 Ω resistor, so $R_N = 12\|12 = 6\,\Omega$.

4.66 Find the Norton equivalent with respect to the terminals a, b for the circuit in Fig. P4.66.





Create a short circuit between the terminals a and b and use mesh current analysis to find $i_N = i_{sc}$. Begin by redrawing the circuit and labeling the mesh currents:

The mesh current equations are

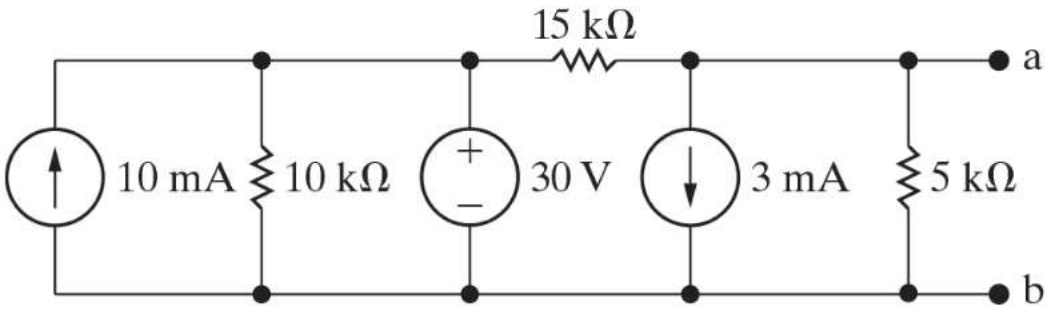
$$25i_1 - 5i_2 - 20i_{sc} = 72;$$

$$-5i_1 + 25i_2 - 8i_{sc} = 0;$$

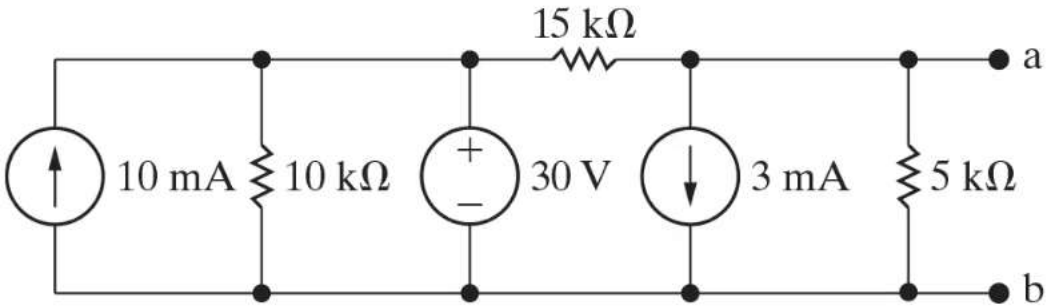
$$-20i_1 - 8i_2 + 28i_{sc} = 0.$$

Solving, $i_1 = 12.72$ A, $i_2 = 6$ A, and $i_N = i_{sc} = 10.8$ A. Therefore, the Norton equivalent circuit is a 10.8 A source in parallel with a 6 Ω resistor.

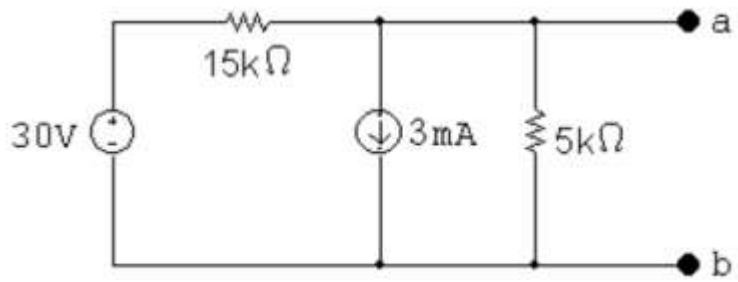
4.67 Find the Norton equivalent with respect to the terminals a, b in the circuit in Fig. P4.67.

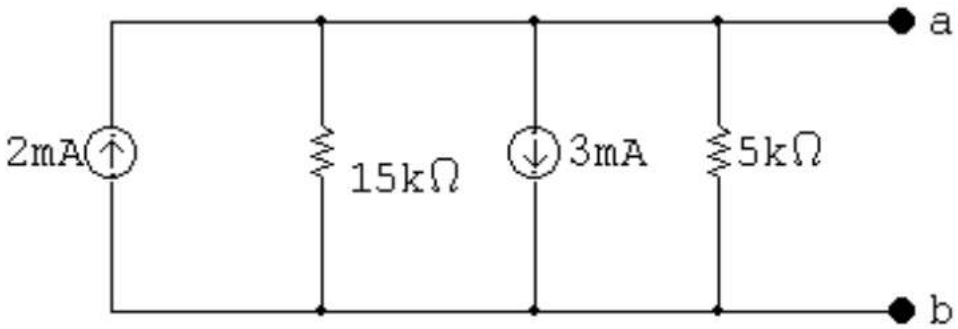


4.67 Find the Norton equivalent with respect to the terminals a, b in the circuit in Fig. P4.67.

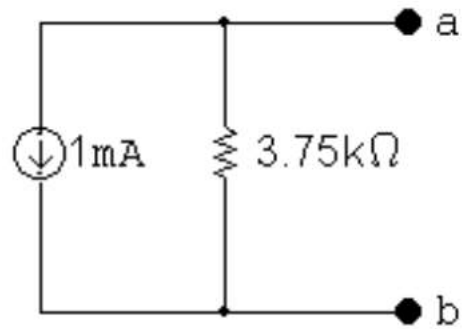


First we make the observation that the 10 mA current source and the 10 kΩ resistor will have no influence on the behavior of the circuit with respect to the terminals a,b. This follows because they are in parallel with an ideal voltage source. Hence our circuit can be simplified to

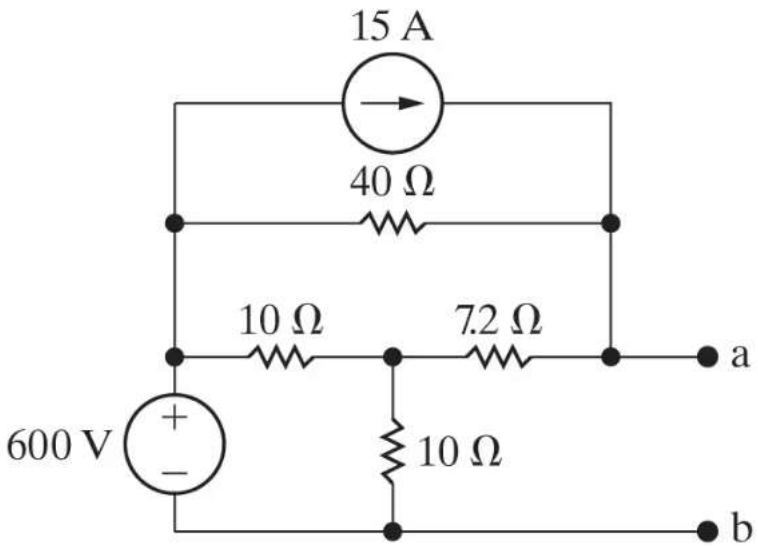




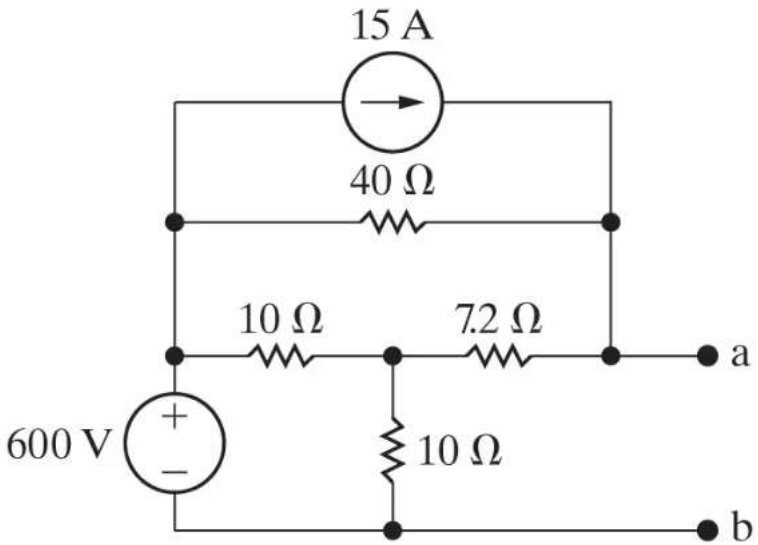
Therefore the Norton equivalent is



4.68 Find the Norton equivalent with respect to the terminals a and b for the circuit in Fig.



4.68 Find the Norton equivalent with respect to the terminals a and b for the circuit in Fig.



(i) $R_{\text{Th}} = \frac{V_{\text{oc}}}{i_{\text{sc}}} \Rightarrow$ Solve V_{oc} by applying mesh currents.

$$600 = 20i_1 - 10i_2 \quad \rightarrow (1)$$

$$600 = 57.2i_2 - 10i_1 \quad \rightarrow (2) \quad \rightarrow i_1 = 39.574 \text{ A}, \quad i_2 = 19.148 \text{ A}$$

$$\Rightarrow V_{\text{oc}} = 10\,(i_1) + 72\,(i_2) = 528.60 \text{ V}$$

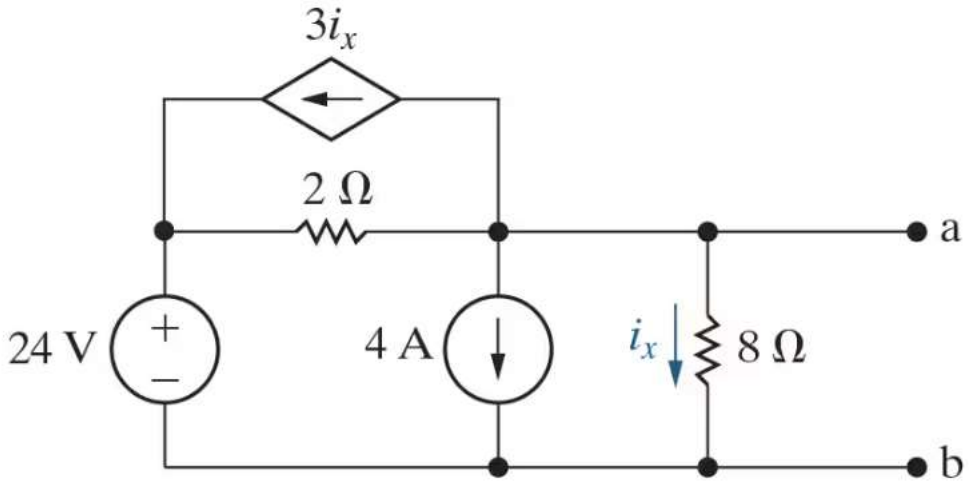
(ii) Solve i_{sc} by applying short circuit across AB and solve mesh currents.

$$600 = 20i_1 - 10i_2 - 10i_{\text{sc}} \rightarrow \quad (1); \qquad 600 = 52i_2 - 10i_1 - 72i_{\text{sc}} \rightarrow \quad (2)$$

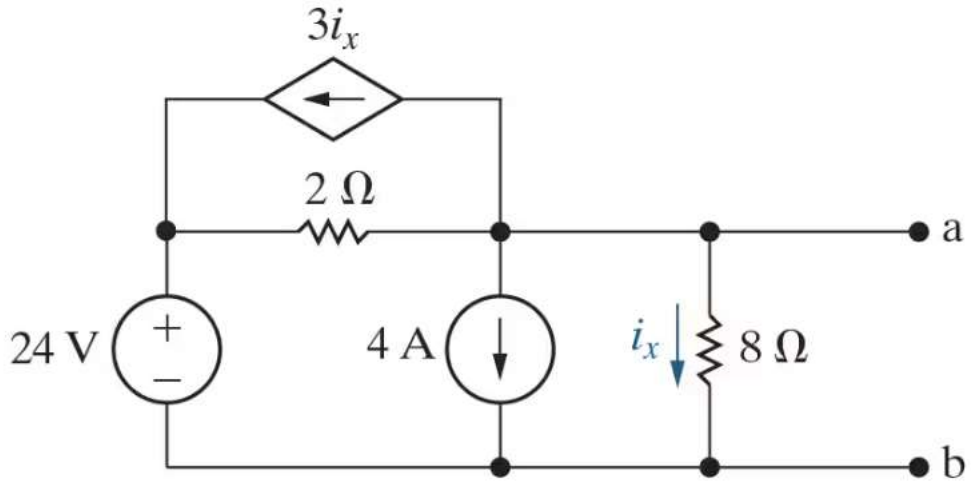
$$17.2i_{\text{sc}} - 10i_1 - 72i_2 = 0 \rightarrow \quad (3)$$

$$\therefore \; i_1 = 78.88 \text{ A}; \quad i_2 = 36.5853 \text{ A}; i_{\text{sc}} = 61.1755 \text{ A}$$

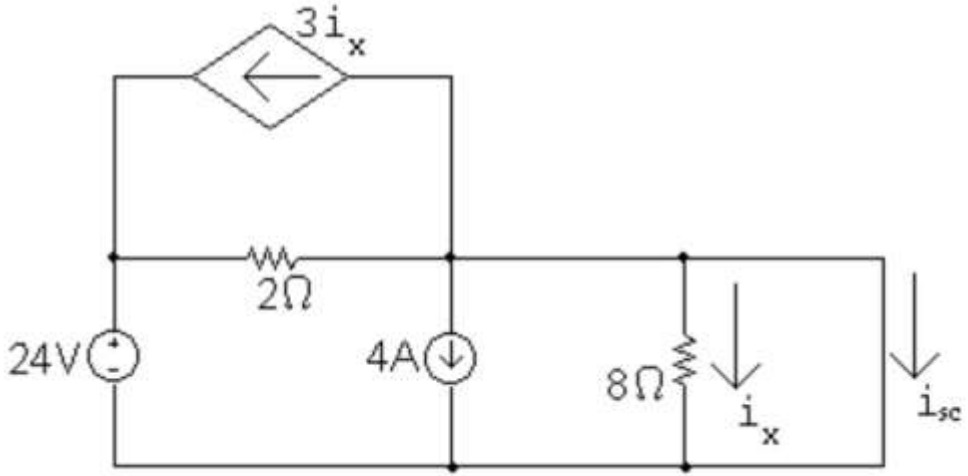
4.75 Find the Norton equivalent with respect to the terminals a, b for the circuit seen in Fig.

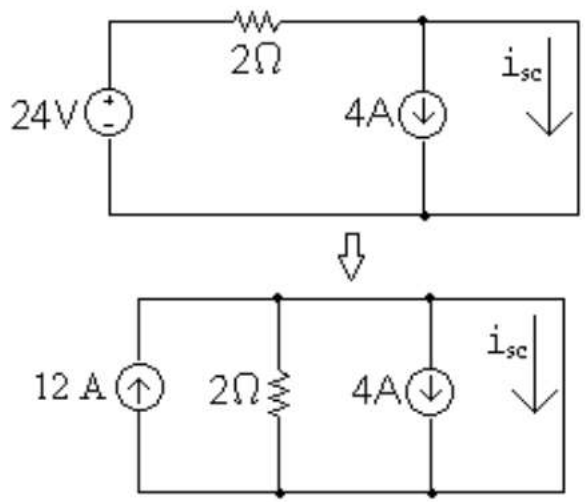


4.75 Find the Norton equivalent with respect to the terminals a, b for the circuit seen in Fig.



Begin by calculating the short circuit current, which is also i_N, from the circuit below:

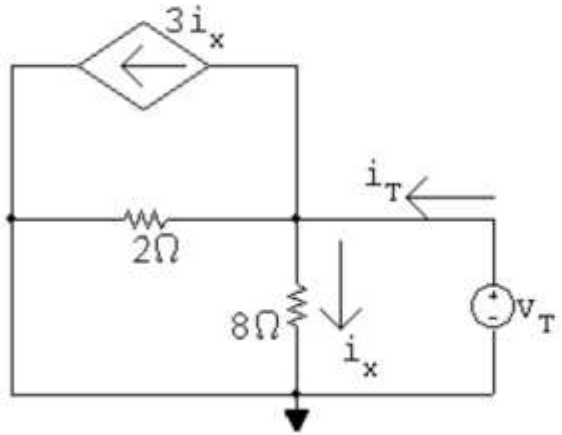




$$i_N = i_{sc} = 12 - 4 = 8\,\text{A}.$$

4.75 Find the Norton equivalent with respect to the terminals a, b for the circuit seen in Fig.

Now calculate R_{Th}. To do this, we use the test source method. Replace the voltage source with a short circuit, the current source with an open circuit, and apply the test voltage v_{T}, as shown in the circuit below:



Write a KCL equation at the middle node:

$$i_{\text{T}} = i_x + 3i_x + v_{\text{T}}/2 = 4i_x + v_{\text{T}}/2.$$

Use Ohm's law to determine i_x as a function of v_{T}:

$$i_x = v_{\text{T}}/8.$$

Substitute the second equation into the first equation:

$$i_{\text{T}} = 4(v_{\text{T}}/8) + v_{\text{T}}/2 = v_{\text{T}}.$$

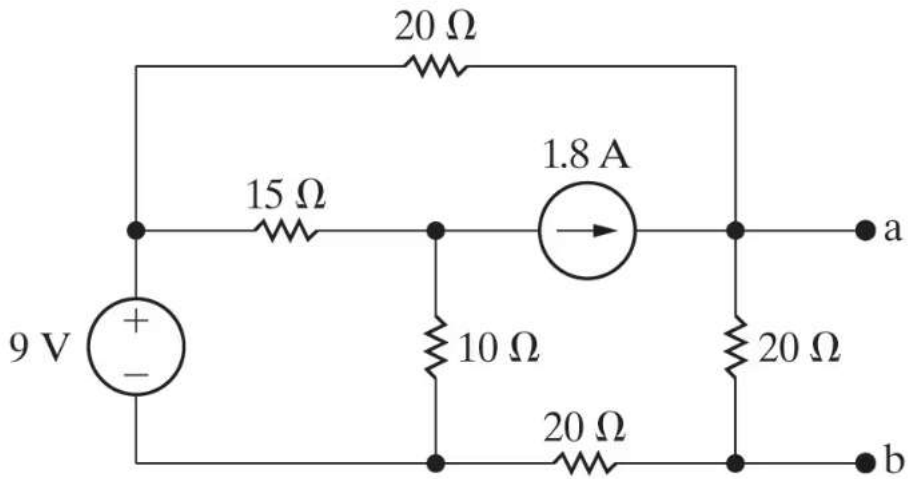
Thus,

$$R_{\text{Th}} = v_{\text{T}}/i_{\text{T}} = 1\,\Omega.$$

The Norton is an 8 A source in parallel with a $1\,\Omega$ resistor.

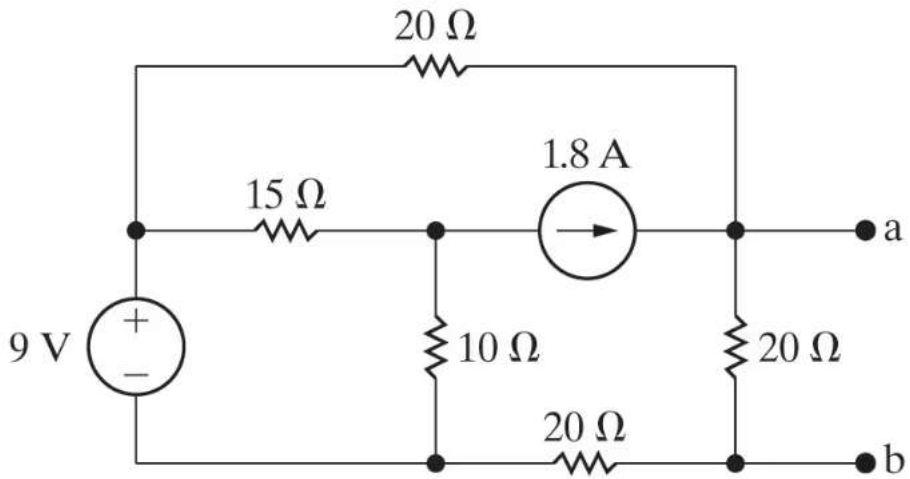
4.77 **a.** Find the Thévenin equivalent with respect to the terminals a, b for the circuit in Fig. P4.77 by finding the open-circuit voltage and short-circuit current.

b. Solve for the Thévenin resistance by removing the independent sources. Compare your result to the Thévenin resistance found in (a).

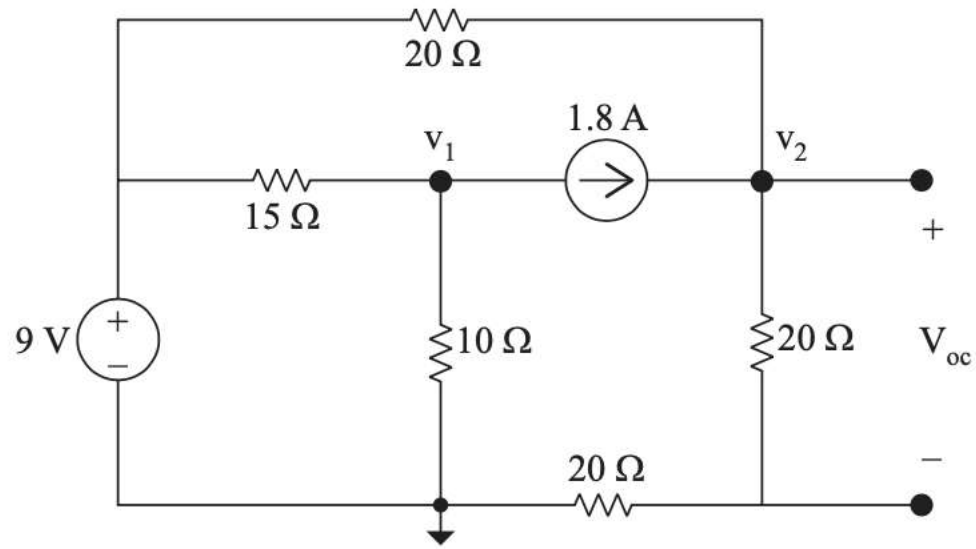


4.77 **a.** Find the Thévenin equivalent with respect to the terminals a, b for the circuit in Fig. P4.77 by finding the open-circuit voltage and short-circuit current.

b. Solve for the Thévenin resistance by removing the independent sources. Compare your result to the Thévenin resistance found in (a).



Open circuit:



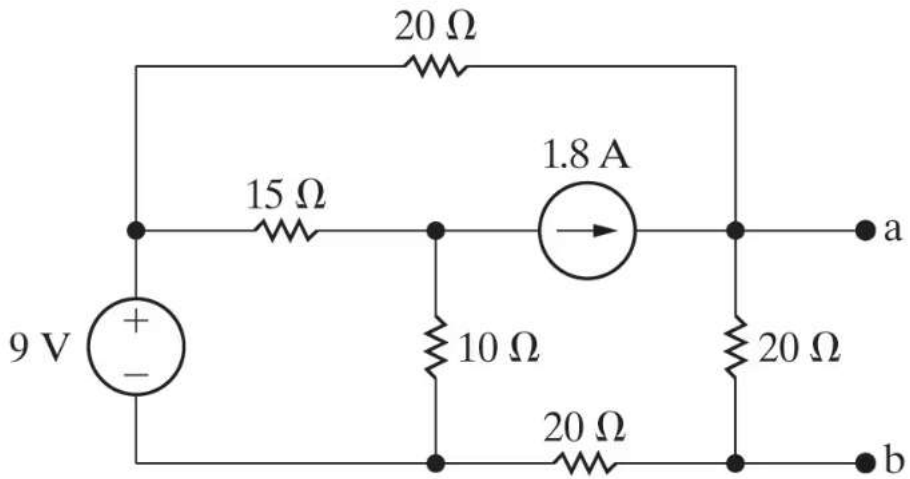
$$\frac{v_2 - 9}{20} + \frac{v_2}{40} - 1.8 = 0;$$

$$v_2 = 30 \text{ V};$$

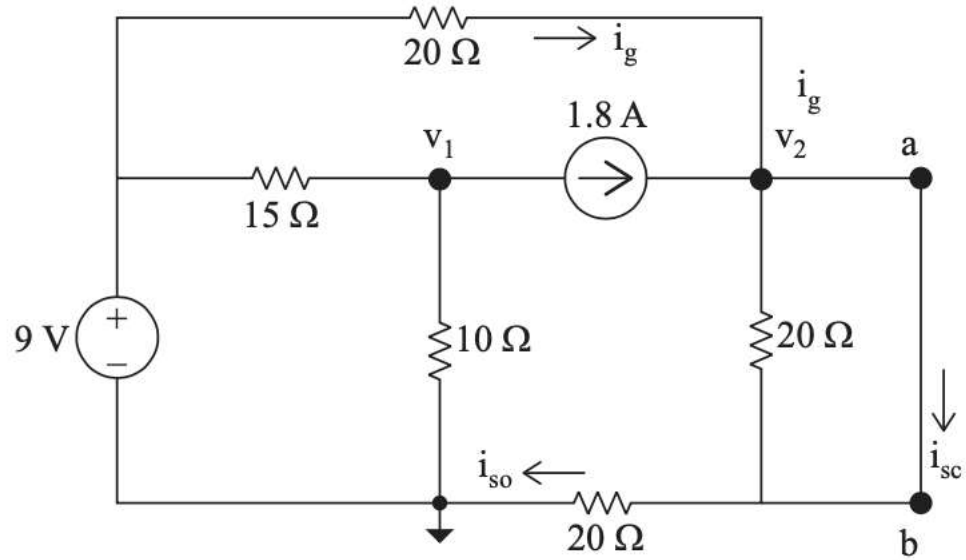
$$v_{\text{Th}} = 30 \times \frac{20}{40} = 15 \text{ V}.$$

4.77 **a.** Find the Thévenin equivalent with respect to the terminals a, b for the circuit in Fig. P4.77 by finding the open-circuit voltage and short-circuit current.

b. Solve for the Thévenin resistance by removing the independent sources. Compare your result to the Thévenin resistance found in (a).



Short circuit:



$$i_a = \frac{9 - 22.5}{20} = -0.675 \text{ A};$$

$$i_{sc} = 1.8 - 0.675 = 1.125 \text{ A};$$

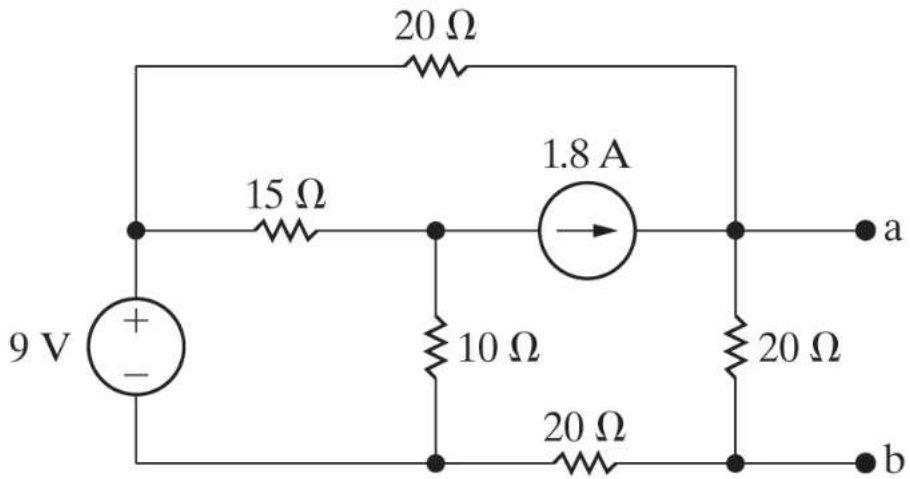
$$R_{Th} = \frac{15}{1.125} = 13.33\ \Omega.$$

$$\frac{v_2 - 9}{20} + \frac{v_2}{20} - 1.8 = 0;$$

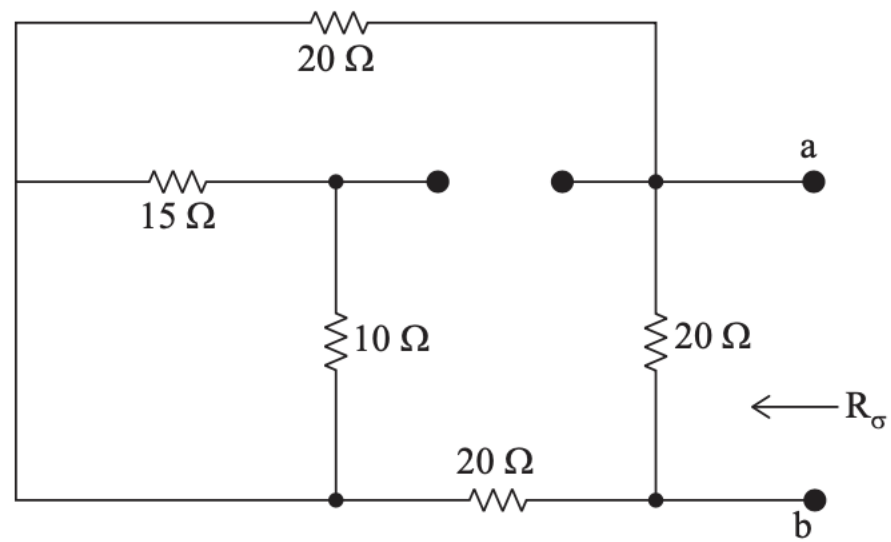
$$\therefore\ v_2 = 22.5 \text{ V}.$$

4.77 **a.** Find the Thévenin equivalent with respect to the terminals a, b for the circuit in Fig. P4.77 by finding the open-circuit voltage and short-circuit current.

b. Solve for the Thévenin resistance by removing the independent sources. Compare your result to the Thévenin resistance found in (a).

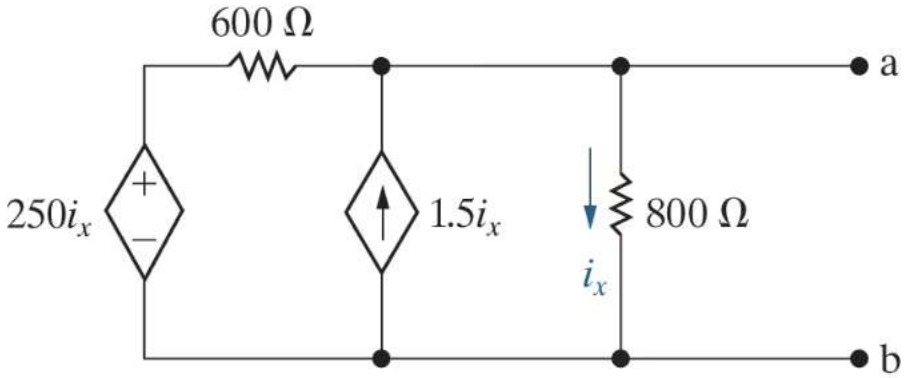


[b]

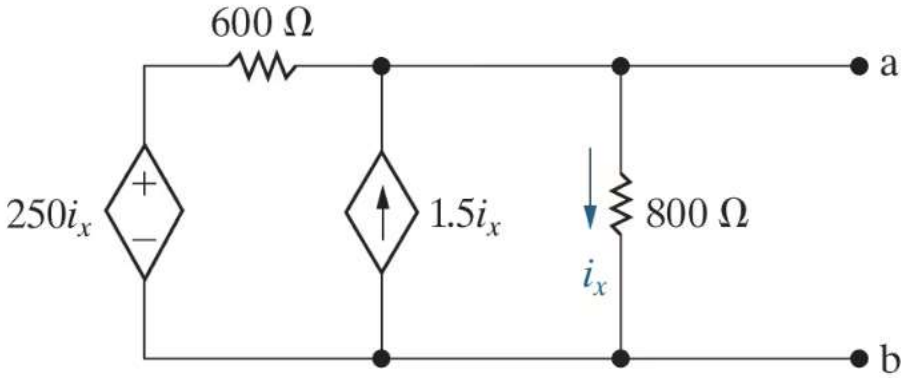


$$R_{\text{Th}} = (20 + 20)20 = 13.33\ \Omega(\text{CHECKS}).$$

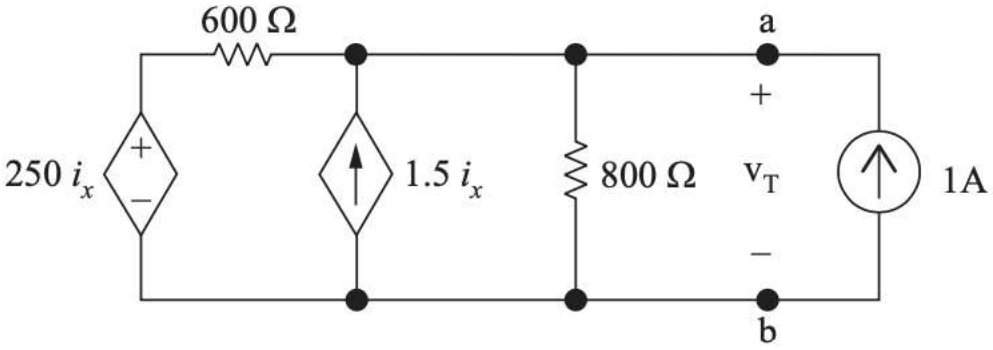
4.79 Find the Thévenin equivalent with respect to the terminals a and b for the circuit in Fig.



4.79 Find the Thévenin equivalent with respect to the terminals a and b for the circuit in Fig.



$V_{\text{Th}} = 0$ since there are no independent sources in the circuit



$$\frac{v - 250i_x}{600} - 1.5i_x + \frac{v}{800} + (-1) = 0$$

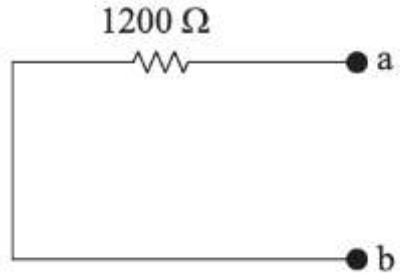
$$i_x = \frac{v}{800}$$

$$\left(1.1458 \times 10^{-3}v\right) - \left(1.875 \times 10^{-3}v\right) + \left(1.25 \times 10^{-3}v\right) - 1 = 0$$

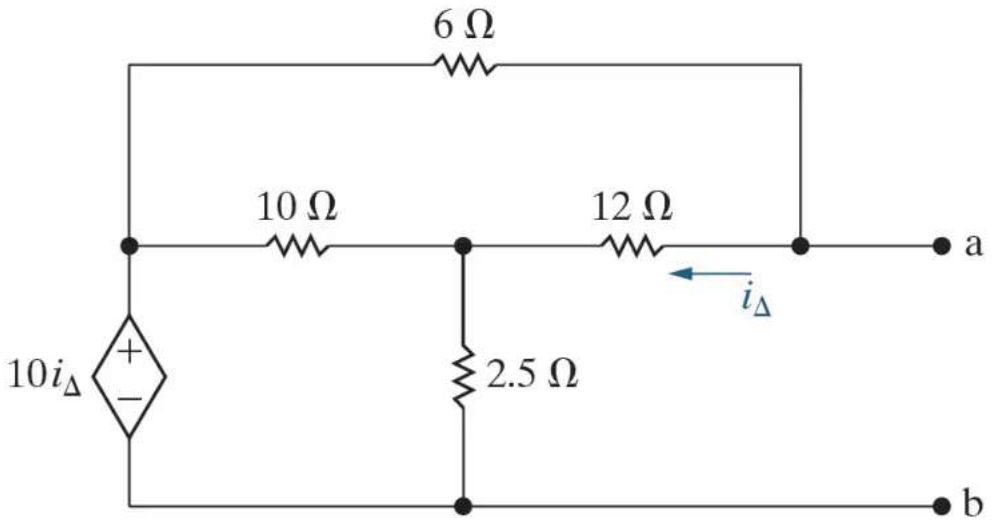
$$8.33 \times 10^{-4}v = 1$$

$$v = 1200.480 \text{ V}; \qquad i_x = 1.5 \text{ A}$$

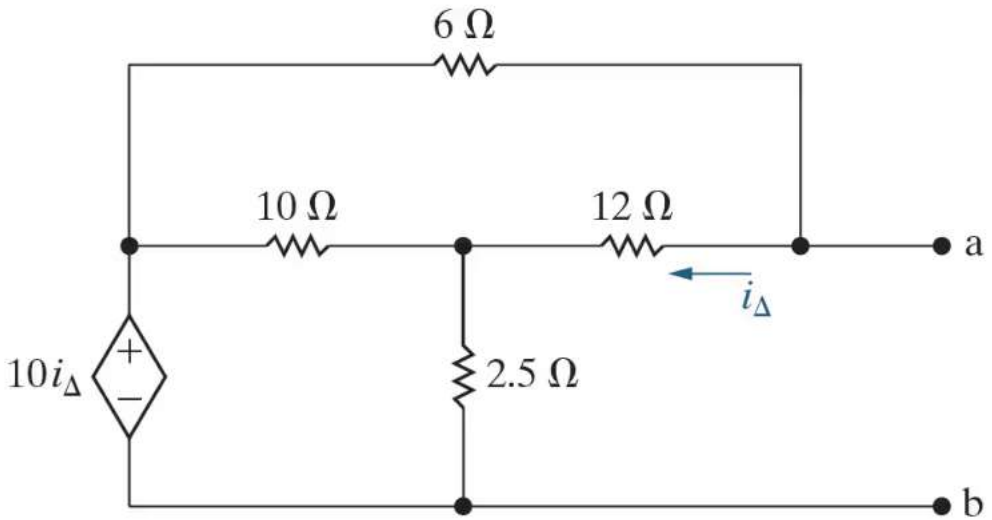
$$R_{\text{Th}} = \frac{v}{1\text{A}} = 1200.48 \approx= 1200\ \Omega$$



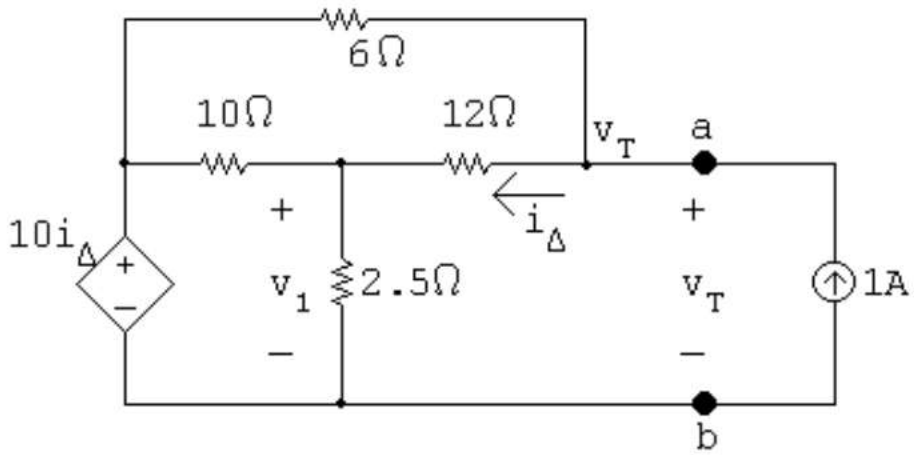
4.80 Find the Thévenin equivalent with respect to the terminals a, b in the circuit in Fig. P4.80.



4.80 Find the Thévenin equivalent with respect to the terminals a, b in the circuit in Fig. P4.80.



$V_{\text{Th}} = 0$, since circuit contains no independent sources.

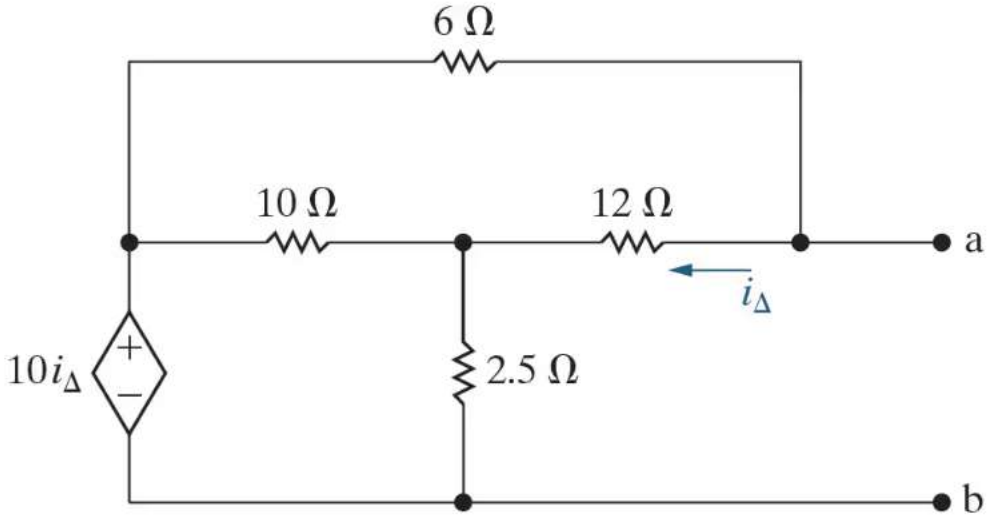


$$\frac{v_1 - 10i_\Delta}{10} + \frac{v_1}{2.5} + \frac{v_1 - v_T}{12} = 0;$$

$$\frac{v_T - v_1}{12} + \frac{v_T - 10i_\Delta}{6} - 1 = 0;$$

$$i_\Delta = \frac{v_T - v_1}{12}.$$

4.80 Find the Thévenin equivalent with respect to the terminals a, b in the circuit in Fig. P4.80.



In standard form:

$$v_1\left(\frac{1}{10}+\frac{1}{2.5}+\frac{1}{12}\right)+v_\mathrm{T}\left(-\frac{1}{12}\right)+i_\Delta(-1)=0;$$

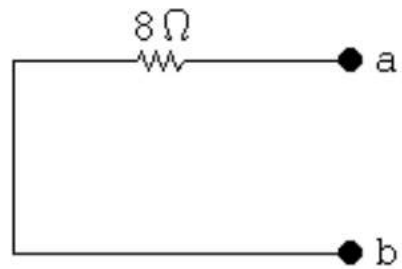
$$v_1\left(-\frac{1}{12}\right)+v_\mathrm{T}\left(\frac{1}{12}+\frac{1}{6}\right)+i_\Delta\left(-\frac{10}{6}\right)=1;$$

$$v_1(1)+v_\mathrm{T}(-1)+i_\Delta(12)=0.$$

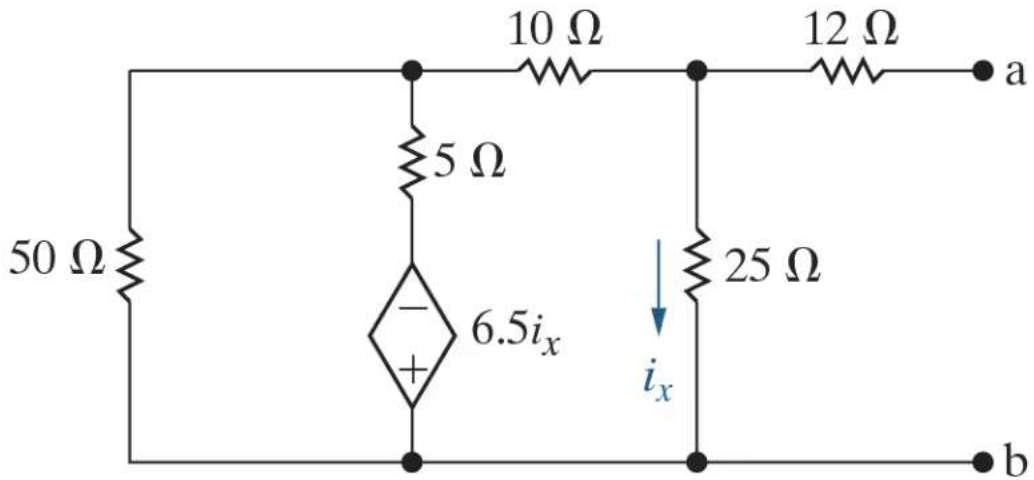
Solving,

$$v_1=2\,\mathrm{V}; \qquad v_\mathrm{T}=8\,\mathrm{V}; \qquad i_\Delta=0.5\,\mathrm{A}.$$

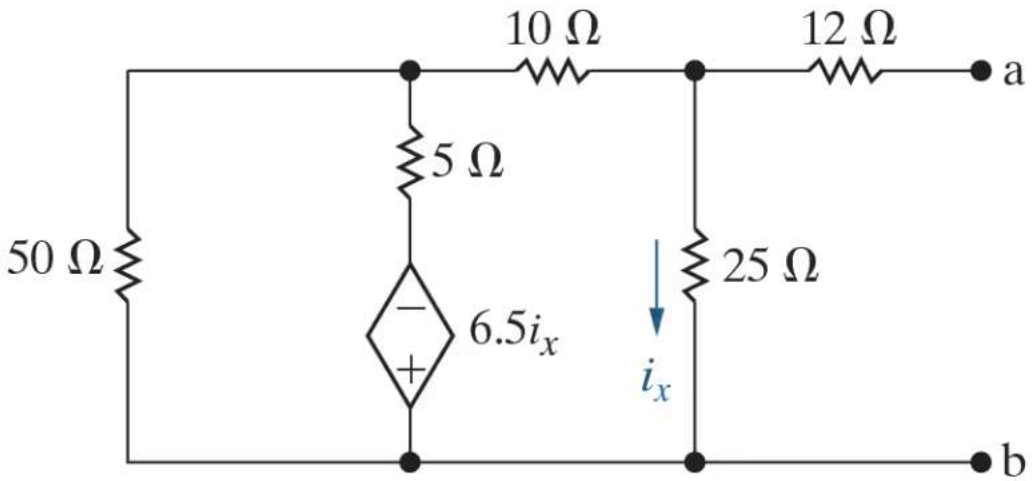
$$\therefore \quad R_\mathrm{Th}=\frac{v_\mathrm{T}}{1\,\mathrm{A}}=8\,\Omega.$$



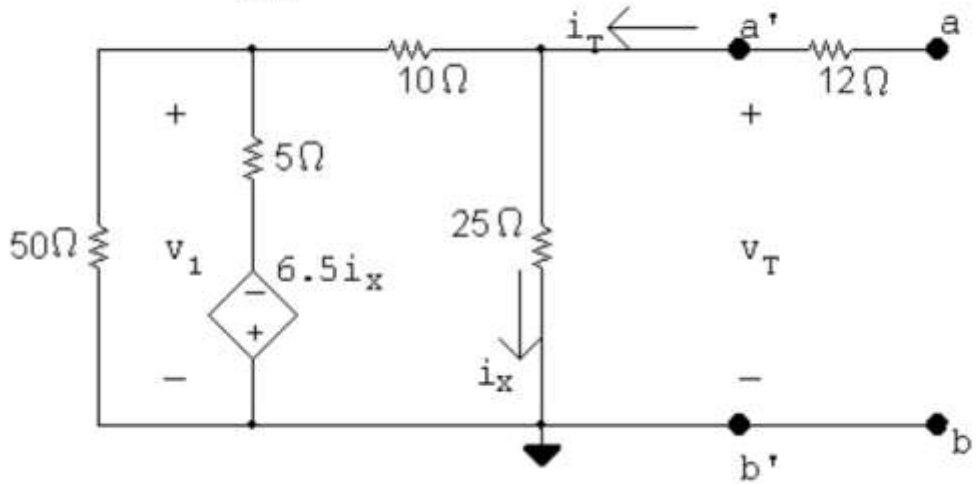
4.81 Find the Thévenin equivalent with respect to the terminals a, b in the circuit in Fig. P4.81.



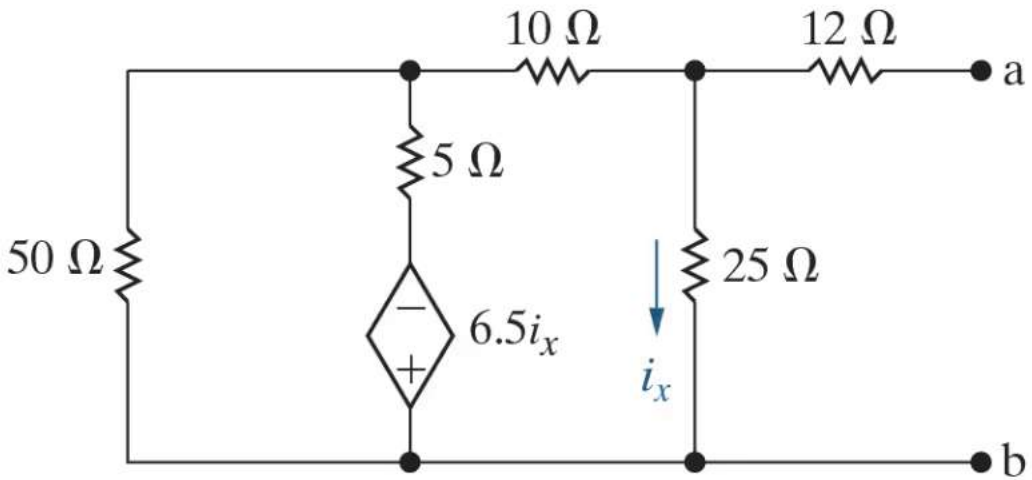
4.81 Find the Thévenin equivalent with respect to the terminals a, b in the circuit in Fig. P4.81.

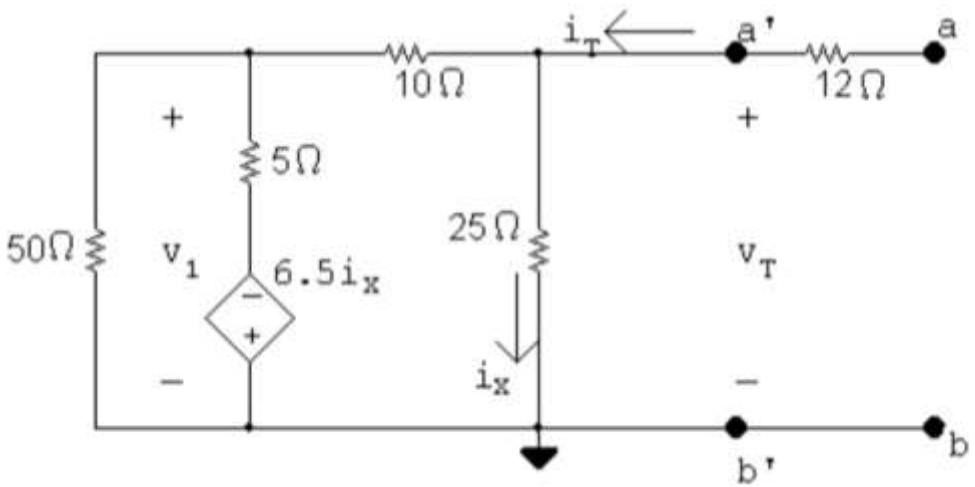


$V_{\text{Th}} = 0$ since there are no independent sources in the circuit. To find R_{Th} we first find $R_{a'b'}$.



4.81 Find the Thévenin equivalent with respect to the terminals a, b in the circuit in Fig. P4.81.





$$i_T = \frac{v_T}{25} + \frac{v_T - v_1}{10};$$

$$\frac{v_1}{50} + \frac{v_1 + 6.5i_x}{5} + \frac{v_1 - v_T}{10} = 0 \text{ so } 16v_1 + 65i_x = 5v_T;$$

$$i_x = \frac{v_T}{25}, \qquad 65i_x = 2.6v_T;$$

$$16v_1 + 2.6v_T = 5v_T;$$

$$\therefore \quad v_1 = 0.15v_T.$$

$$i_T = \frac{v_T}{25} + \frac{v_T - 0.15v_T}{10} = \frac{6.25}{50}v_T;$$

$$\frac{v_T}{i_T} = 50/6.25 = 8\,\Omega = R_{a'b'};$$

$$\therefore \quad R_{Th} = 12 + 8 = 20\,\Omega.$$

